

However, there are several shortcomings to this method. First, one has to verify that Φ is a bijection. In addition, to succeed, this method requires some regularity of the boundary of Ω . Moreover, one is still faced with the question of solving the Dirichlet problem for Ω . At this stage Riemann proposed using the “Dirichlet principle.” But applying this idea involves difficulties that must be overcome.⁵

Nevertheless, using different methods, one can prove the existence of the mapping in the general case. This approach is carried out below in Section 3.

2 The Schwarz lemma; automorphisms of the disc and upper half-plane

The statement and proof of the Schwarz lemma are both simple, but the applications of this result are far-reaching. We recall that a rotation is a map of the form $z \mapsto cz$ with $|c| = 1$, namely $c = e^{i\theta}$, where $\theta \in \mathbb{R}$ is called the angle of rotation and is well-defined up to an integer multiple of 2π .

Lemma 2.1 *Let $f : \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(0) = 0$. Then*

- (i) $|f(z)| \leq |z|$ for all $z \in \mathbb{D}$.
- (ii) If for some $z_0 \neq 0$ we have $|f(z_0)| = |z_0|$, then f is a rotation.
- (iii) $|f'(0)| \leq 1$, and if equality holds, then f is a rotation.

Proof. We first expand f in a power series centered at 0 and convergent in all of $\mathbb{D}$

$$f(z) = a_0 + a_1z + a_2z^2 + \cdots.$$

Since $f(0) = 0$ we have $a_0 = 0$, and therefore $f(z)/z$ is holomorphic in $\mathbb{D}$ (since it has a removable singularity at 0). If $|z| = r < 1$, then since $|f(z)| \leq 1$ we have

$$\left| \frac{f(z)}{z} \right| \leq \frac{1}{r},$$

and by the maximum modulus principle, we can conclude that this is true whenever $|z| \leq r$. Letting $r \rightarrow 1$ gives the first result.

For (ii), we see that $f(z)/z$ attains its maximum in the interior of $\mathbb{D}$ and must therefore be constant, say $f(z) = cz$. Evaluating this expression

⁵An implementation of Dirichlet’s principle in the present two-dimensional situation is taken up in Book III.

at z_0 and taking absolute values, we find that $|c| = 1$. Therefore, there exists $\theta \in \mathbb{R}$ such that $c = e^{i\theta}$, and that explains why f is a rotation.

Finally, observe that if $g(z) = f(z)/z$, then $|g(z)| \leq 1$ throughout $\mathbb{D}$, and moreover

$$g(0) = \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z} = f'(0).$$

Hence, if $|f'(0)| = 1$, then $|g(0)| = 1$, and by the maximum principle g is constant, which implies $f(z) = cz$ with $|c| = 1$.

Our first application of this lemma is to the determination of the automorphisms of the disc.

2.1 Automorphisms of the disc

A conformal map from an open set Ω to *itself* is called an **automorphism** of Ω . The set of all automorphisms of Ω is denoted by $\text{Aut}(\Omega)$, and carries the structure of a group. The group operation is composition of maps, the identity element is the map $z \mapsto z$, and the inverses are simply the inverse functions. It is clear that if f and g are automorphisms of Ω , then $f \circ g$ is also an automorphism, and in fact, its inverse is given by

$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}.$$

As mentioned above, the identity map is always an automorphism. We can give other more interesting automorphisms of the unit disc. Obviously, any rotation by an angle $\theta \in \mathbb{R}$, that is, $r_\theta : z \mapsto e^{i\theta}z$, is an automorphism of the unit disc whose inverse is the rotation by the angle $-\theta$, that is, $r_{-\theta} : z \mapsto e^{-i\theta}z$. More interesting, are the automorphisms of the form

$$\psi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha}z}, \quad \text{where } \alpha \in \mathbb{C} \text{ with } |\alpha| < 1.$$

These mappings, which were introduced in Exercise 7 of Chapter 1, appear in a number of problems in complex analysis because of their many useful properties. The proof that they are automorphisms of $\mathbb{D}$ is quite simple. First, observe that since $|\alpha| < 1$, the map ψ_α is holomorphic in the unit disc. If $|z| = 1$ then $z = e^{i\theta}$ and

$$\psi_\alpha(e^{i\theta}) = \frac{\alpha - e^{i\theta}}{e^{i\theta}(e^{-i\theta} - \bar{\alpha})} = e^{-i\theta} \frac{w}{\bar{w}},$$

where $w = \alpha - e^{i\theta}$, therefore $|\psi_\alpha(z)| = 1$. By the maximum modulus principle, we conclude that $|\psi_\alpha(z)| < 1$ for all $z \in \mathbb{D}$. Finally we make

the following very simple observation:

$$\begin{aligned}
 (\psi_\alpha \circ \psi_\alpha)(z) &= \frac{\alpha - \frac{\alpha-z}{1-\bar{\alpha}z}}{1 - \bar{\alpha} \frac{\alpha-z}{1-\bar{\alpha}z}} \\
 &= \frac{\alpha - |\alpha|^2 z - \alpha + z}{1 - \bar{\alpha}z - |\alpha|^2 + \bar{\alpha}z} \\
 &= \frac{(1 - |\alpha|^2)z}{1 - |\alpha|^2} \\
 &= z,
 \end{aligned}$$

from which we conclude that ψ_α is its own inverse! Another important property of ψ_α is that it vanishes at $z = \alpha$; moreover it interchanges 0 and α , namely

$$\psi_\alpha(0) = \alpha \quad \text{and} \quad \psi_\alpha(\alpha) = 0.$$

The next theorem says that the rotations combined with the maps ψ_α exhaust all the automorphisms of the disc.

Theorem 2.2 *If f is an automorphism of the disc, then there exist $\theta \in \mathbb{R}$ and $\alpha \in \mathbb{D}$ such that*

$$f(z) = e^{i\theta} \frac{\alpha - z}{1 - \bar{\alpha}z}.$$

Proof. Since f is an automorphism of the disc, there exists a unique complex number $\alpha \in \mathbb{D}$ such that $f(\alpha) = 0$. Now we consider the automorphism g defined by $g = f \circ \psi_\alpha$. Then $g(0) = 0$, and the Schwarz lemma gives

$$(2) \quad |g(z)| \leq |z| \quad \text{for all } z \in \mathbb{D}.$$

Moreover, $g^{-1}(0) = 0$, so applying the Schwarz lemma to g^{-1} , we find that

$$|g^{-1}(w)| \leq |w| \quad \text{for all } w \in \mathbb{D}.$$

Using this last inequality for $w = g(z)$ for each $z \in \mathbb{D}$ gives

$$(3) \quad |z| \leq |g(z)| \quad \text{for all } z \in \mathbb{D}.$$

Combining (2) and (3) we find that $|g(z)| = |z|$ for all $z \in \mathbb{D}$, and by the Schwarz lemma we conclude that $g(z) = e^{i\theta}z$ for some $\theta \in \mathbb{R}$. Replacing z by $\psi_\alpha(z)$ and using the fact that $(\psi_\alpha \circ \psi_\alpha)(z) = z$, we deduce that $f(z) = e^{i\theta}\psi_\alpha(z)$, as claimed.

Setting $\alpha = 0$ in the theorem yields the following result.

Corollary 2.3 *The only automorphisms of the unit disc that fix the origin are the rotations.*

Note that by the use of the mappings ψ_α , we can see that the group of automorphisms of the disc acts **transitively**, in the sense that given any pair of points α and β in the disc, there is an automorphism ψ mapping α to β . One such ψ is given by $\psi = \psi_\beta \circ \psi_\alpha$.

The explicit formulas for the automorphisms of $\mathbb{D}$ give a good description of the group $\text{Aut}(\mathbb{D})$. In fact, this group of automorphisms is “almost” isomorphic to a group of 2×2 matrices with complex entries often denoted by $\text{SU}(1, 1)$. This group consists of all 2×2 matrices that preserve the hermitian form on $\mathbb{C}^2 \times \mathbb{C}^2$ defined by

$$\langle Z, W \rangle = z_1 \bar{w}_1 - z_2 \bar{w}_2,$$

where $Z = (z_1, z_2)$ and $W = (w_1, w_2)$. For more information about this subject, we refer the reader to Problem 4.

2.2 Automorphisms of the upper half-plane

Our knowledge of the automorphisms of $\mathbb{D}$ together with the conformal map $F : \mathbb{H} \rightarrow \mathbb{D}$ found in Section 1.1 allow us to determine the group of automorphisms of $\mathbb{H}$ which we denote by $\text{Aut}(\mathbb{H})$.

Consider the map

$$\Gamma : \text{Aut}(\mathbb{D}) \rightarrow \text{Aut}(\mathbb{H})$$

given by “conjugation by F ”:

$$\Gamma(\varphi) = F^{-1} \circ \varphi \circ F.$$

It is clear that $\Gamma(\varphi)$ is an automorphism of $\mathbb{H}$ whenever φ is an automorphism of $\mathbb{D}$, and Γ is a bijection whose inverse is given by $\Gamma^{-1}(\psi) = F \circ \psi \circ F^{-1}$. In fact, we prove more, namely that Γ preserves the operations on the corresponding groups of automorphisms. Indeed, suppose that $\varphi_1, \varphi_2 \in \text{Aut}(\mathbb{D})$. Since $F \circ F^{-1}$ is the identity on $\mathbb{D}$ we find that

$$\begin{aligned} \Gamma(\varphi_1 \circ \varphi_2) &= F^{-1} \circ \varphi_1 \circ \varphi_2 \circ F \\ &= F^{-1} \circ \varphi_1 \circ F \circ F^{-1} \circ \varphi_2 \circ F \\ &= \Gamma(\varphi_1) \circ \Gamma(\varphi_2). \end{aligned}$$

The conclusion is that the two groups $\text{Aut}(\mathbb{D})$ and $\text{Aut}(\mathbb{H})$ are the same, since Γ defines an isomorphism between them. We are still left with the